

UAS PDF PIN Mailer Implementation Guide - 6.0.1-GA (AI)

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1. Preface - UAS PDF PIN Mailer Implementation Guide

This document provides the concept, implementation steps and API integration details of the AccessMatrix Universal Authentication Server (UAS) PDF PIN Mailer solution with different Hardware Security Modules (HSMs).

Intended Reader

This is intended as a reference for the security administrator, professional service and developers.

Document Scope

While this document provides necessary information and steps to implement the UAS PDF PIN mailer with HSM, it will not provide documentation on operating the HSM.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- [AccessMatrix Common Administration Guide](#)
- [UAS Administration Guide](#)
- [UAS E2EEA Developer Guide](#)
 - [UAS E2EEA with SafeNet ProtectServer HSM Implementation](#)
 - [UAS E2EEA with SafeNet Payment HSM Implementation](#)
 - [UAS E2EEA with Thales payShield 10K HSM Implementation](#)
 - [UAS E2EEA with Utimaco CryptoServer HSM Implementation](#)
 - [UAS Pseudo E2EEA Implementation](#)

2. Introduction to UAS PDF PIN Mailer Implementation

This document shows you how to implement and integrate the UAS PDF PIN mailer with HSM to secure the login secrets for internet services and deliver sensitive PDF files such as pay advice slips.

Terminology

This table lists some common terms that are used in this document. It is not intended as a comprehensive definition for each item.

Term	Description
End-to-End Encryption Authentication (E2EEA)	The AccessMatrix UAS End-to-End Encryption Authentication (E2EEA) solution. End-to-End Encryption (E2EE) creates a security domain between the customer's terminal and the Hardware Security Module (HSM service provider). It allows the administrator to integrate with HSM and front-end component without any complex code for password encryption during the user's login.
Self-Service Facility (SSF)	The Self-Service Facility (SSF) is an interface that allows users to create a beneficiary, entitlement request, business role request, approvals and provisioning of requests.
Simple Mail Transfer Protocol (SMTP)	An SMTP server is a mail server or machine used to send and receive mail messages that take care of the whole email delivery process.

About AccessMatrix UAS PDF PIN Mailer

The AccessMatrix UAS PDF PIN mailer solution enables organizations to deliver sensitive PIN mailers to end-users, simplifying integration complexities.

The solution covers the entire PIN mailer generation and delivery:

- Data merging and encrypted PDF generation
- PIN generation
- Integration with AccessMatrix UAS End to End Encryption Authentication (E2EEA) Module for PIN protection, verification, and storage

The following workflow shows the implementation of AccessMatrix UAS PDF PIN mailer. The PIN verification is using E2EEA with the respective HSM you use.

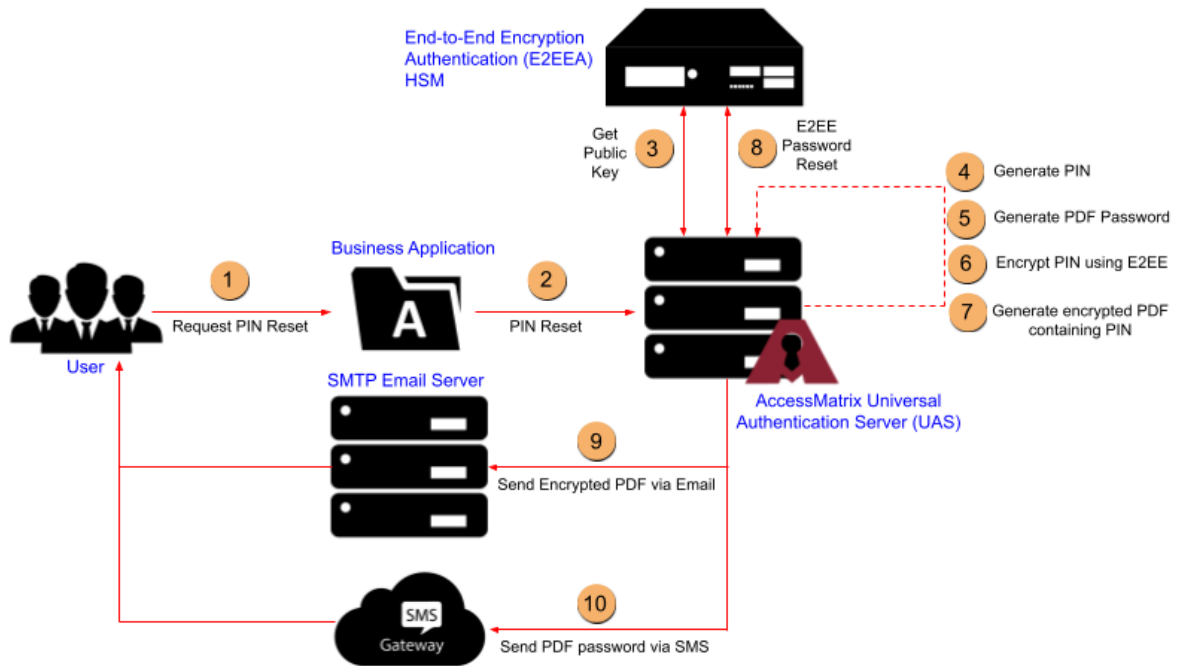
1. User requests PIN to be reset from the target business application.
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Note: The user is assumed to be authenticated via suitable means by the front-end application for Self-Service Facility (SSF).

2. The PIN has been reset and sent to AccessMatrix server for authentication.
3. Next, AccessMatrix server will obtain a public key from the respective HSM for encryption.
4. The public key PIN will utilize the AccessMatrix server to generate a secret pin.
5. AccessMatrix UAS will also generate the PDF password.
6. Next, it encrypts the PIN using E2EE.
7. It generates an encrypted PDF that contains a secret PIN.
8. It sends the PDF containing the secret PIN to the HSM E2EE to reset the password.
9. Next, for the typical use case where the PDF PIN and password are sent to the user:
 - a. The AccessMatrix server sends the encrypted PDF to the SMTP email server to be sent to the user.
 - b. The AccessMatrix server sends the PDF password to the SMS gateway to be sent to the user.
10. Alternatively, for the use case where the business requires the PDF PIN files to be saved locally for printing and mailing to users:
 - a. The AccessMatrix server saves the encrypted PDF to the specified local path.
 - b. The AccessMatrix server returns the PDF password in the response to the Reset Password API call.
 - c. After the PDF PIN mailer has been retrieved and printed, the business deletes the files for security purposes.

The diagram below illustrates an implementation of the AccessMatrix UAS PDF PIN mailer using email delivery.



Supported Hardware Security Modules (HSMs)

The AccessMatrix UAS PDF PIN mailer supports the following HSMs.

- SafeNet Payment (PHEFT) HSM
- SafeNet ProtectServer HSM
- Thales payShield 10K HSM
- Utimaco CryptoServer HSM
- Pseudo-E2EEA HSM

Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer

There are two types of workflow use in AccessMatrix UAS to improve security and eliminate the possibility of one person controlling the PIN mailer:

- Single-Control PIN Mailer
- Dual-Control PIN Mailer



Note: Only the administrator can approve and trigger the Print, Delete and Withdraw actions.

The supported HSMs for AccessMatrix UAS PDF PIN mailer support these two types of PIN mailer workflow are listed below.

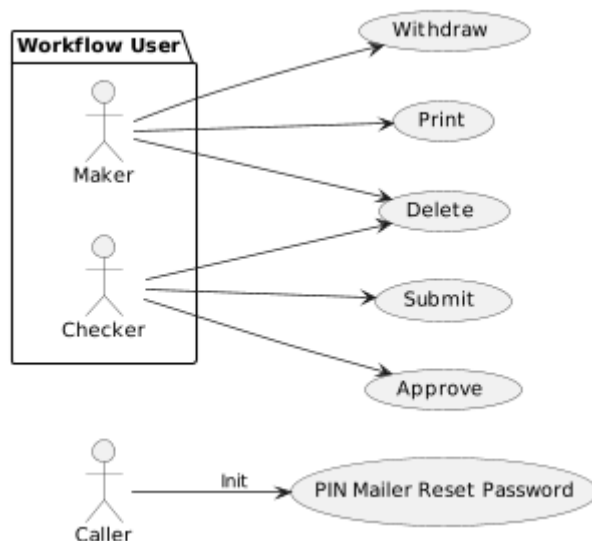
- SafeNet Payment (PHEFT) HSM
- SafeNet ProtectServer HSM
- Thales payShield 10K HSM
- Utimaco CryptoServer HSM
- Pseudo-E2EEA HSM

These workflows involve a **caller**, a **maker** and a **checker**. See their roles below.

- A **caller** is a user who submits a PIN mailer Reset Password request in the On-Site WorkForce (OSWF); this caller is the UUID of the AccessMatrix user who triggers the password reset.
- A **maker** is any user who has access to the Admin Console with administrator privileges for the PIN mailer view and processes. In a typical scenario, this user triggers the **Print** action. The maker can immediately delete or withdraw a caller's request without requiring a checker's approval in a single-control PIN mailer workflow.

i **Note:** In the Dual-Control workflow, a checker must approve before triggering the **Print** and **Delete** actions.

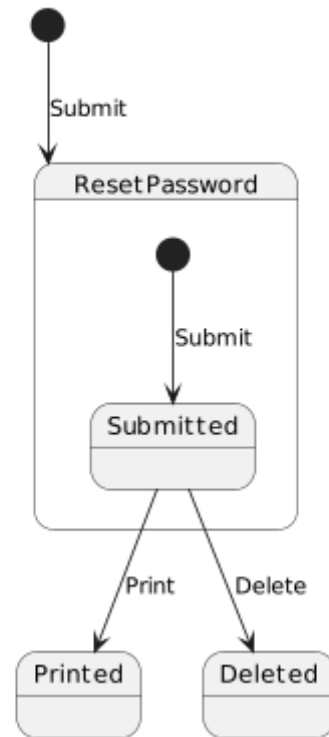
- A **checker** is any user who has access to the Admin Console with administrator privileges for the PIN mailer view and processes who should not be a maker. For example, in a typical scenario, this user approves or deletes the actions triggered by a maker.



Single-Control PIN Mailer Workflow

This workflow involves a caller and a maker only, and the **Print** and **Delete** actions do not require a checker's approval to trigger.

Single-Control PIN Mailer State Diagram



Common Use Cases

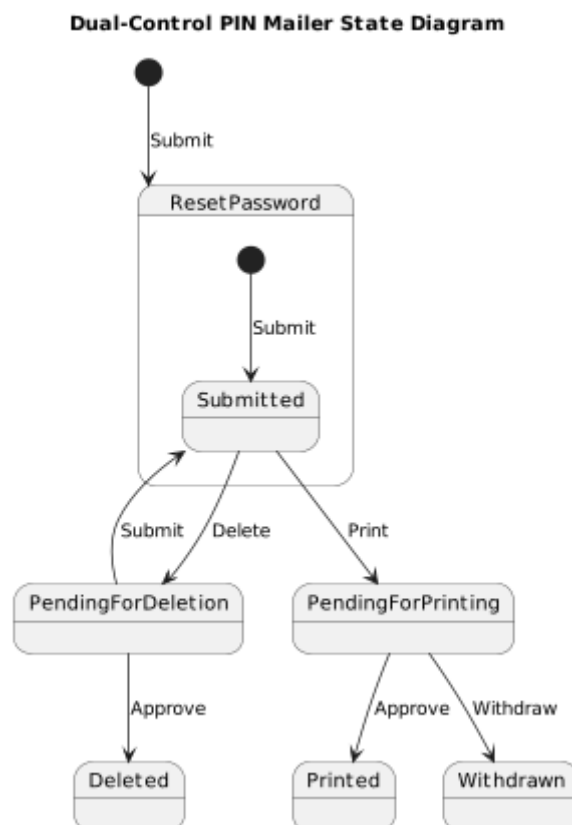
- After a user (caller) resets the password, the maker triggers the **Print** action.
 1. A caller initiates the reset password API (**PDF System Assigned Pending**).
Status: Submitted.
 2. A maker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Print** to trigger the print action.
Status: Printed.
- After a user (caller) resets the password, the maker triggers the **Delete** action.
 1. A caller initiates the reset password API (**PDF System Assigned Pending**).
Status: Submitted.

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2. A maker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Delete** to trigger the delete action.

Status: Deleted.

Dual-Control PIN Mailer Workflow

This workflow involves a caller, a maker and a checker to approve the **Print** and **Delete** actions triggered by a maker.



Common Use Cases

- After resetting the password, the maker triggers the **Print** action; the printing will be carried out after the checker approves.
 1. A caller triggers the reset password API (PDF System Assigned Pending).
Status: Submitted.
 2. A maker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Print** to trigger the print action.
Status: Pending printing.
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3. A checker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Approve** to trigger the approve action.
Status: Printed.
- After resetting the password, the maker triggers the **Print** action, and the checker deletes the request.
 1. A caller triggers the reset password API (PDF System Assigned Pending).
Status: Submitted.
 2. A maker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Print** to trigger the print action.
Status: Pending printing.
 3. A checker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests, and clicks the button **Delete** to trigger the delete action.
Status: Deleted.
 - After resetting the password, the maker triggers **Print** action, but before the checker approves the request, the maker withdraws the request (The maker must view the request, withdraw from the list doesn't work.).
 1. Repeat steps 1 and 2 as shown above for the roles of the caller and the maker.
 2. The maker logs in to the Admin Console and navigates to **Request > PIN Mailer**, selects one request (maker must go into and view the request, and clicks the button **Withdraw** to trigger the withdraw action).
Status: Withdrawn.
 - After resetting the password, the maker triggers the **Delete** action, and the checker approves or re-submit the request.
 1. Repeat steps 1 and 2 as shown above for the roles of the caller and the maker.
 2. These will lead to either of the following 2 actions:
 - The checker logs in to Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests and clicks the button **Approve** to trigger the delete action.
Status: Deleted
 - The checker logs in to Admin Console and navigates to **Request > PIN Mailer**, selects one or more requests and clicks the button **Submit** to trigger the submit
-

action.

Status: Submitted and start from the first flow.

2.1. Implementation Planning

Below is the checklist for implementing the UAS PDF PIN mailer solution.

Tasks	Reference
Secure AccessMatrix Server License	License
Configure the PDF PIN Mailer Message Template	PDF PIN Mailer Message Template
Configure the PDF Delivery Message Template	PDF Delivery Message Template
Configure the PDF Password Message Template	PDF Password Message Template
Configure the PDF PIN Mailer Delivery Channel	PDF PIN Mailer Delivery Channel
Configure the PDF Password Delivery Channel	PDF Password Delivery Channel
Create the PDF Signing Key	PDF Signing Key
Configure the Login Module	Login Module
Create the Authentication Realm	Configure Authentication Realm
Assign Authentication Realm to User	Assign Authentication Realm to User



Note: Throughout the configuration steps, you will note that some variables are used to specify the user's name, email and others. For Internal users, use the variables "\$targetUserName"; "\$targetUserEmail". For external users, use the variables "\${event.targetUserName}"; "\${event.targetUserEmail}".

Implementation Planning

Planning prior to environment setup is critical to the success of UAS PDF PIN mailer implementation. Understanding the customer's nature of business, studying the customer's requirements and existing environments will help identify the actions to make before the actual implementation.

Security Consideration

The PIN mailer is insecure; therefore, it relies on the respective HSM to generate the random numbers and prevent disclosure of the PIN across transactions or the delivery channels by ensuring the customer is the only person who can view their PINs online. UAS PDF PIN mailer is used to issue PINs via PDF and HSM in a secure way to prevent attacks that reveal the PINs. This is widely used in organizations for login

secrets of internet services and pay advice slips. The customers will receive their PINs that are embedded inside an encrypted PDF document.

3. UAS PDF PIN Mailer Configuration

Before doing any PDF PIN mailer-related configurations, check that:

- AccessMatrix UAS E2EEA solution has already been implemented successfully.
- The respective HSM you are using has been installed and is working properly.

Setup Steps

This section defines the steps relating to the implementation of the UAS PDF PIN mailer service. Before you start the implementation, it is assumed that you have completed the planning and implementation checklists and you have understood the concept of the UAS PDF PIN mailer solution.

- **Upload the License**

Server license is used to enable general and product-specific features. To enable PDF PIN mailer support in AccessMatrix server, refer to [Upload License](#) for the details.

- **Configure the PDF PIN Mailer Message Template**

This template defines the layout and alignment of the recipient name, password, date, etc., in the PDF file. Refer to [Configure PDF PIN Mailer Message Template](#) for the configuration details.

- **Configure the PDF Delivery Message Template**

This template defines the message (e.g. an email message) containing the PIN to send to the recipient. Refer to [Configure PDF Delivery Message Template](#) for the configuration details.

- **Configure the PDF Password Message Template**

This template defines the message (e.g. an SMS message) containing the PDF password to send to the recipient. This password is required to open the PDF PIN mailer. Refer to [Configure PDF Password Message Template](#) for the configuration details.

- **Configure the PDF PIN Mailer Delivery Channel**

This delivery channel is used to send the PDF mail notification to the recipient, e.g. using the SMTP (Simple Mail Transfer Protocol) server via SMTP protocol to deliver the message. Refer to [Configure PDF PIN Mailer Delivery Channel](#) for the configuration details.

- **Configure PDF Password Delivery Channel**

This delivery channel is used to send the PDF password notification to the recipient, e.g. using HTTP (HyperText Transfer Protocol) server via HTTP protocol to deliver the message in SMS (Short Message Service). Refer to [Configure PDF Password Delivery Channel](#) for the configuration details.



Note: For security reason, use different channels for PDF password notification and PDF PIN mailer delivery channel.

- **Create a Digital Signature for PDF PIN Mailer**

Digital signatures are meant to assure authenticity and data integrity of digital documents, to verify that the document originator is who they claim to be and to ascertain that a PDF digital document has not been tampered with between the moment of signing and the subsequent viewing. You use a certificate and a private key to sign the PDF PIN mailer. This section shows you how to create a PDF signing key. Refer to [Create PDF Signing Key](#) for the configuration details.

- **Configure the Login Module**

Login module provides a login or authentication method to verify user login credentials. Refer to [Configure Login Module](#) for the configuration details.

- **Configure the Authentication Realm**

The authentication realm associates user credentials for authentication based on a lookup module and login module(s) where user entries can be created into the local repository of AccessMatrix or integrated from LDAP (Lightweight Directory Access Protocol). Refer to [Configure Authentication Realm](#) for the details.

- **Assign Login Accounts to the User**

User is an entity that accesses the system and may belong to a default store (local repository) or an external user store, e.g. AD (Active Directory). Assign the configured authentication realm to the user by adding it to the **Login Accounts** tab. Refer to [Assign Authentication Realm to User](#) for the details.

Upload License

Before doing any configuration in the Admin Console, secure the server licence required for PDF PIN mailer support. Ensure that **E2EEA Pin Mailer** and **E2EEA Base** license features are set to "Yes" to enable PDF PIN mailer support. These settings will also display the required login module for the HSM in the login module implementation list.

To load the server license:

1. Navigate to **Administration Dashboard > System > Server & Cache** and click **Server** on the left pane.
 2. Select the Server Id and click **Edit**.
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3. Select the **Licensed Features** tab and select the first radio button. Browse for the license file.
 4. Click **Upload Licence File** to upload the license file to the server.
 5. Save the changes.
 6. Restart the AccessMatrix server for the new license to take effect.

Configure PDF PIN Mailer Message Template

To create the PDF PIN mailer template:

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Message Template**.
2. Click **Create** and select the "Coordinate-based Template" implementation.
3. Click **Next** button.
4. Enter a **Message Template Id** (e.g. "PDFPinMailerTemplate") and a **Message Template Name** (e.g. "PDFPinMailerTemplate").
5. Specify the summary of the message in **Delivery Summary**.
6. Specify the values for the **Body** attribute. This contains the message body's content and indicates the position and text info to be embedded in the PDF attachment. Example:

Row	Column	Text
156	110	\${user.name}
212	210	<<E2EEPIN>>



Note: The macro <<E2EEPIN>> for the PIN must be a single entry in the template and not have any other characters around it.

7. Upload the file via the **Pin Mailer Template In PDF** attribute. The customer should provide this PDF file, and it contains the empty form/template of the PDF file. Refer to [Appendix A](#) for a sample PDF file.
8. Click **Save**. A new PDF PIN mailer message template is created.

Refer to [AccessMatrix Common Administration Guide - Configure Message Template](#) for detailed information about the Message Template attributes.

Configure PDF Delivery Message Template

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Message Template**.
2. Click **Create** and select the "Default Template Engine" implementation.
3. Click **Next** button.
4. Enter a **Message Template Id** (e.g. "PDFDeliveryTemplate") and a **Message Template Name** (e.g. "PDFDeliveryTemplate").
5. Fill in the necessary values for the following attributes:
 - a. **From** - sender of the message, e.g. "customercare@bank.com.sg"
 - b. **To** - primary recipient(s) of the message, e.g. "\${targetUserEmail}"
 - c. **Subject** - the short summary of the message, e.g. "UAS Reset Password PIN"
 - d. **Body** - contains the message body's content. Example:
"Please find the attached file, which contains your PIN."
The PDF file is encrypted and requires your PDF password to be sent separately."
6. Set **Allow File Attachment** to "true" to allow PDF attachment during email delivery.
7. Click **Save**. A new PDF delivery message template is created.

Refer to [AccessMatrix Common Administration Guide - Configure Message Template](#) for detailed information about the message template attributes.

Configure PDF Password Message Template

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Message Template**.
 2. Click **Create** and select the "Default Template Engine" implementation.
 3. Click **Next** button.
 4. Enter a **Message Template Id** (e.g. "PDFPwdTemplate") and a **Message Template Name** (e.g. "PDFPwdTemplate").
 5. Fill in the necessary values for the following attributes:
 - a. **From** - sender of the message, e.g. "customercare@bank.com.sg"
 - b. **To** - primary recipient(s) of the message, e.g. "\${targetUserMobile}"
-

-
- c. **Subject** - the short summary of the message, e.g. 'UAS PDF Password'
 - d. **Body** - contains the message body's content. Example:
"Dear \${targetUserName}, please be informed that the PDF password \${pdfpassword}
for
PIN Mailer has been created."
6. Click **Save**. A new PDF password message template is created.

Refer to [AccessMatrix Common Administration Guide - Configure Message Template](#) for detailed information about the message template attributes.

Configure PDF PIN Mailer Delivery Channel

Email Deliverer

The steps below explain how to configure the PDF PIN mailer delivery channel to send the PDF PIN via email:

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Delivery Channel**.
2. Click **Create** and select the "Email Deliverer" implementation from the "**Choose Delivery Channel Implementation**" page.
3. Click **Next**.
4. Enter a **Delivery Channel Id** (e.g. "EmailDelivery") and a **Delivery Channel Name** (e.g. "EmailDelivery").
5. Specify the values for the delivery channel attributes.
6. Click **Save**. A new email delivery channel is created.

Refer to [AccessMatrix Common Administration Guide - Configure Delivery Channel](#) for detailed information about the email deliverer attributes.

File Deliverer

You can use the file deliverer delivery channel if you intend to save the generated PDF PIN files to a local folder for printing.

The steps below explain how to configure the PDF PIN mailer delivery channel to save the encrypted PDF PIN files to a local folder:

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Delivery Channel**.
2. Click **Create** and select the "File Deliverer" implementation from the **Choose Delivery Channel Implementation** page.
3. Click **Next**.
4. Enter a **Delivery Channel Id** (e.g. "FileDelivery") and a **Delivery Channel Name** (e.g. "FileDelivery").
5. Specify the installation folder that the PDF PIN file will be saved to under the **Destination Folder** attribute.
6. Enter "pdfFileName" under the **File Name Attribute** attribute.
7. Click **Save**. A new file delivery channel is created.

 **Notes:**

- Generated PDF files are saved with the following naming conventions:
 - If the user is not created in a segment, **<userId>,<Timestamp>.pdf**.
 - If the user is created in a segment, **<SegmentId>,<userId>,<Timestamp>.pdf**.
For example, the generated PDF file may be named **demouser,20220810142225.pdf**.
- To open the saved PDF Pin files, the PDF password is still required. Refer to the [Configure PDF Password Delivery Channel](#) section for steps on how to configure the PDF password to be returned directly to the API caller via API response.
- Refer to [AccessMatrix Common Administration Guide - Configure Delivery Channel](#) for detailed information about the File Deliverer attributes.

Configure Separate PDF PIN Mailer Validity for File Deliverer

 **Note:** Skip this section if the scenario described below does not apply to your deployment.

The validity of the PDF PIN mailer is typically determined by the **PDF Pin Mailer Validity Period (days)** attribute under the relevant E2EEA login module. However, if you need to support both the Email Deliverer and File Deliverer delivery channels in the same deployment, it may be necessary to configure

a different validity period for each delivery channel. For example, a longer validity period may be needed for File Deliverer PDF PIN mailers that will be printed and mailed to end-users.

The following steps provide instructions on configuring a separate validity period for File Deliverer PDF PIN mailers. Once this has been configured, the validity period specified under the **PDF Pin Mailer Validity Period (days)** attribute will no longer apply to File Deliverer PDF PIN mailers and will be overridden.

1. Navigate to `<AM5_ROOT>/tomcat/webapps/am5/WEB-INF/classes/` and open the **wpe_config.properties** file.
2. Enable the **Scripted Rules** advanced feature in the admin console. To do so, first search for the `advancedFeatures.enable` element and change the associated value to "true":

Properties (.properties files)
<code>advancedFeatures.enable =true</code>

3. Add the line `loginModule.scriptedRules = true` to the file:

Properties (.properties files)
<code>advancedFeatures.enable =true</code> <code>loginModule.scriptedRules =true</code>

4. Save the changes and restart the AccessMatrix server.
5. Open the Admin Console and navigate to **Administration Dashboard > Configuration > Authentication > Login Module**. Select the relevant E2EEA login module.
6. Under the newly added **Scripted Rules** tab, add the following script to the text box:

Code
<pre>if (event.isResetPasswordGenerationOptionSystemPDF() && "testFileDeliverer".equals(event.getParam("overrideDeliveryChannel"))) { //Valid for 24 hours action.setLoginAccountEndTimeInHours(24); } //Clear expiry upon user changing password during his/her 1st login if (event.isAfterChangePassword() && event.getLoginAccountEndTime()!=null) { action.setLoginAccountEndTime(null); }</pre>

-
7. Set the validity period of the File Deliverer PDF PIN by entering an appropriate value (in hours) for `action.setLoginAccountEndTimeInHours()`. For example, enter `action.setLoginAccountEndTimeInHours(24)` for 24 hours.
 8. Save the changes.

Configure PDF Password Delivery Channel

PDF Password Delivery via SMS

The steps below explain to configure the PDF password delivery channel to send the PDF password via SMS.



Note: You can also have the PDF password sent to the user's email inbox by selecting the "Email Deliverer" implementation instead.

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > Delivery Channel**.
2. Click **Create** and select the "HTTP Deliverer" implementation from the **Choose Delivery Channel Implementation** page.
3. Click **Next**.
4. Enter a **Delivery Channel Id** (e.g. "SMSDelivery") and a **Delivery Channel Name** (e.g. "SMSDelivery").
5. Specify the values for the delivery channel attributes.
6. Click **Save**. A new SMS delivery channel is created.

Refer to [AccessMatrix Common Administration Guide - Configure Delivery Channel](#) for detailed information about the HTTP Deliverer attributes.

PDF Password Delivery via API Response

The PDF password can be configured to be returned to the Password Reset REST API caller. This is to facilitate the printing of PIN mailers.



Note: This is only supported for the SafeNet ProtectServer E2EEA and Utimaco CryptoServer E2EEA login modules.

To configure this:

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1. Navigate to **Administration Dashboard > Configuration > Authentication > Login Module** and select the relevant login module.
 2. Under the **PDF PIN Mailer** tab, set the **Allow Send PDF Password in Response** attribute to "true".

i **Note:** If the **Allow Send PDF Password in Response** attribute is set to "false", the PDF password will be delivered via the delivery channel specified in the **PDF Password Delivery Channel** attribute.

3. Click **Save**.

Refer to [Configure SafeNet ProtectServer E2EEA Login Module](#) and [Configure Utimaco CryptoServer E2EEA Login Module](#) for detailed information about the respective login module attributes.

Create PDF Signing Key

A PDF signing Key can be created through the following:

- [Imported Key Store \(Digital Signature\)](#)
- [PKCS11 Key Provider](#)

Imported Key Store (Digital Signature)

i **Note:** The key store or digital signature consists of a digital trusted certificate and a private key. You can use any 3rd party application\authority to configure a key store by setting up a CA certificate and a key pair (private and public key).

1. Navigate to **Administration Dashboard > Security Policy > Key Management > Encryption Key**.
2. Click **Create** and select "SystemKeyProvider" from the **Key Provider** drop-down list and then select "Upload Java keystore for PDF Pin Mailer Signing" from the **Encryption key template** drop-down list.
3. Click **Next**.
4. Enter an **Encryption Key id** (i.e. "Key01") and an **Encryption Key Name** (i.e. "PDFSigningKey").
5. Enter the key's description (i.e. "PDF signing key for PIN mailer").
6. Select **Choose File** to select a keystore (JKS) file you have generated and stored in your PC.

! **Important Note:** AccessMatrix UAS supports Java keystore only.

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7. Enter the alias of the certificate, the keystore passphrase, and the private key passphrase.
 8. Select "Data Integrity" from the **Purpose** drop-down list.
 9. Select the required values on the **Storage and Key Wrapping** and **Access Control** tabs.
 10. Click **Save**. A PDF signing key is created and saved to the database of AccessMatrix UAS.
 11. (OPTIONAL) You can click **Back** to view the newly created PDF signing key, or export these details to an external keystore using either a different or the same certificate alias, key store passphrase and private key passphrase for other purposes.
 12. Next, add the PDF signing key to the **PDF Pin Mailer** tab when configuring the login module.

PKCS11 Key Provider

A HSM can be used as a key provider to generate the encryption key used for PDF signing.

Perform the following steps:

1. Navigate to **Administration Dashboard > Security Policy > Key Management > Key Provider**.
 2. Click **Create** and select "PKCS11 Key Provider" from the **Implementation** drop-down list.
 3. Click **Next**.
 4. Enter the **Key Provider Id** (i.e. "KeyProv01") and the **Key Provider Name** (i.e. "PDFSigningKeyProv").
 5. Enter the Key Provider's description (i.e. "PDF signing key provider for PIN mailer").
 6. Specify the file path for the ***PKCS11 configuration** file attribute.
 7. Specify the remaining attributes as necessary and click **Save**. The Key Provider has been created.
 8. Navigate to the **Administration Dashboard > Security Policy > Key Management > Encryption Key**.
 9. Click **Create** and select the the previously Key Provider created from the **Key Provider** drop-down list. Next, select "Asymmetric key (key pair)" from the **Encryption key template** drop-down list.
 10. Click **Next**.
 11. Enter an **Encryption Key id** (i.e. "Key01") and an **Encryption Key Name** (i.e. "PDFSigningKey").
 12. Enter the key's description (i.e. "PDF signing key for PIN mailer").
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13. Select "Data Integrity" from the **Purpose** drop-down list.
 14. Select the required values on the **Storage and Key Wrapping** and **Access Control** tabs.
 15. Click **Save**. A PDF signing key is created and saved to the database of AccessMatrix UAS.
 16. Next, add the PDF signing key to the **PDF Pin Mailer** tab when configuring the login module.

Configure Login Module

This section describes the configuration required in the login module to support the PDF PIN mailer service. Refer to the login module configurations according to the respective HSM you use:

Configure Pseudo E2EEA Login Module

In supporting E2EEA with PDF PIN mailer for Pseudo HSM, you must create a new login module. This login module provides the Pseudo E2EEA login method for the user who may reside in either an internal or external user store.

To create a Pseudo E2EEA login module:

1. Navigate to **Administration Dashboard > Configuration > Authentication** and click **Login Module** on the left pane.
2. Click the **Create** button and select the "Pseudo E2EEA Login Module" implementation from the **Choose Login Module Implementation** page.
3. Then, click **Next** and provide the necessary values to the **Configuration** tab attributes (refer to [UAS E2EEA Developer Guide - UAS Pseudo E2EEA Implementation](#) for the configuration details).
4. Select the **PDF Pin Mailer** tab and provide the necessary values for the PIN mailer attributes before saving the changes.

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Tab			
Reset Password Options			
Allow Reset By System Assigned PDF	Set to "true" to automatically generate the password in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within	Y	N

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	UAS. Default: false		
Allow Reset By System Assigned PDF (Pending)	Set to "true" to reset the password using PIN mailer workflow, where upon completion, the password is automatically generated in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
PDF Pin Mailer Generation Configuration			
PDF Signing Key	Use a certificate with a private key to sign PDF. New keys can be created at Administration Dashboard > Security Policy > Key Management > Encryption Key.	N	N
PDF Pin Mailer Validity Period (days)	This value is used to calculate the expiry date of the PDF PIN mailer, where PDF PIN mailer expiry date = creation date of the PDF PIN Mailer + PDF Pin Mailer Validity Period. Note: To show the expiry date in PDF, set <<expiryDate>> macro in the PDF Pin Mailer Message Template.	Y	N
PDF Pin Mailer Expiry Date Format	Date format of the PDF PIN mailer expiry date. Symbol Meaning [Example] y year [e.g. yyyy for 2012] M month in year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9] d day in month [e.g. dd for 10] z time zone [e.g. Z for PST] Example: to display expiry date 10/11/2025 in PDF, enter dd/MM/yyyy.	Y	
Message Template Configuration			
PDF Pin Mailer	Message template of the PDF PIN mailer ("Coordinate-based Template") which allows you to put information like a PIN	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Message Template	value in an encrypted PDF. It can be created at Administration Dashboard > Configuration > Messaging & Response, select "Coordinate-based Template" for the implementation. Set the following in the message template body: • <<E2EEPIN>> to specify the PIN value in PDF • \${pinmailer.xxx} to specify attribute passed from API (using pinmailer.attr.xxx in the params) • \${userAttr.email} to specify the email of the user if the user has its email attribute stored or mapped in AccessMatrix user store (e.g. LDAP, defaultstore or JDBC user store).		
PDF Delivery Message Template	Message template to include the PDF PIN mailer; this template allows you to attach the PDF file in email. It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to attach a PDF file (i.e. "Default Template Engine"). You must set Allow File Attachment to "true" in the message template.	Y	
PDF Password Message Template	Message template of PDF Password allows you to send the PDF password (PDF password is required to open the encrypted PDF file). It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to send a text (i.e. "Default Template Engine"). Set the following in the message template: • \${pdfpassword} in the body field to specify the PDF password.	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	• <code>\${targetUserEmail}</code> in the recipient field to specify user's email. • <code>\${targetUserMobile}</code> in the recipient field to specify the user's mobile.		
Delivery Channel Configuration			
PDF Pin Mailer Delivery Channel	Delivery channel of PDF PIN mailer. Default: Email	Y	
PDF Password Delivery Channel	Delivery channel of PDF password. For security reasons, this channel should be different from PDF PIN Mailer Delivery Channel . Default: SMS	Y	
Pin Mailer Workflow			
PDF Pin Mailer Workflow Type	Refer to Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer . It supports Single-Control and Dual-Control PIN mailers. Select the type of PIN mailer workflow from the drop-down list. Default: Dual-Control PIN Mailer	Y	This setting is mandatory provided the Allow Reset by System Assigned PDF (Pending) option is allowed.

Configure SafeNet PHEFT Login Module

In supporting E2EEA with PDF PIN mailer for SafeNet Payment HSM, you must create a new login module or edit the default SafeNet PHEFT E2EE login module ("SafeNetPHEFTE2EE"). This login module provides the SafeNet PHEFT login method for the user who may reside in either an internal or external user store.

To create a SafeNet PHEFT login module:

1. Navigate to **Administration Dashboard > Configuration > Authentication** and click login module on the left pane.
2. Click the **Create** button and select the "SafeNet PHEFT Login" implementation from the **Choose Login Module Implementation** page.

3. Then, click **Next** and provide the necessary values to the **Configuration** tab attributes (refer to [UAS E2EEA Developer Guide - UAS E2EEA with SafeNet Payment HSM Implementation](#) for the configuration details).
4. Select the **PDF Pin Mailer** tab and provide the necessary values for the PIN mailer attributes before saving the changes.

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Tab			
Reset Password Options			
Allow Reset By System Assigned PDF	Set to "true" to automatically generate the password in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
Allow Reset By System Assigned PDF (Pending)	Set to "true" to reset the password using PIN mailer workflow, where upon completion, the password is automatically generated in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
PDF Pin Mailer Generation Configuration			
PDF Pin Mailer Generation Option	Options to generate the PDF PIN Mailer. There are 2 options available. "Disabled" indicates that PDF PIN mailer generation is turned off. "AccessMatrix" is the embedded option to generate the PDF PIN mailer by AccessMatrix server. Default: Disabled	Y	
PDF Signing Key	Use a certificate with a private key to sign PDF. New keys can be created at Administration Dashboard > Security Policy > Key Management > Encryption Key.	N	N

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Validity Period (days)	This value is used to calculate the expiry date of the PDF PIN mailer where PDF mailer expiry date = creation date of the PDF PIN mailer + PDF Pin Mailer Validity Period. Note: To show the expiry date in PDF, set <<expiryDate>> macro in the PDF Pin Mailer Message Template.	Y	N
PDF Pin Mailer Expiry Date Format	Date format of the PDF PIN mailer expiry date. Symbol Meaning [Example] y year [e.g. yyyy for 2012] M month in year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9] d day in month [e.g. dd for 10] z time zone [e.g. Z for PST] Example: to display expiry date 10/11/2025 in PDF, enter dd/MM/yyyy.	Y	
Message Template Configuration			
PDF Pin Mailer Message Template	Message template of the PDF PIN mailer ("Coordinate-based Template") which allows you to put information like a PIN value in an encrypted PDF. It can be created at Administration Dashboard > Configuration > Messaging & Response, select "Coordinate-based Template" for the implementation. Set the following in the message template body: • <<E2EEPIN>> to specify the PIN value in PDF • \${pinmailer.xxx} to specify attribute passed from API(using pinmailer.attr.xxx in the params) • \${userAttr.email} to specify the email of the user if the user has its email attribute stored or mapped in AccessMatrix user store (e.g. LDAP, defaultstore or JDBC user store).	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Delivery Message Template	Message template to include the PDF PIN mailer; this template allows you to attach the PDF file in email. It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to attach a PDF file (i.e. "Default Template Engine"). You must set Allow File Attachment to "true" in the message template.	Y	
PDF Password Message Template	Message template of PDF Password allows you to send the PDF password (PDF password is required to open the encrypted PDF file). It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to send a text (i.e. "Default Template Engine"). Set the following in the message template: • \${pdfpassword} in the body field to specify the PDF password. • \${targetUserEmail} in the recipient field to specify user's email. • \${targetUserMobile} in the recipient field to specify the user's mobile.	Y	
Delivery Channel Configuration			
PDF Pin Mailer Delivery Channel	Delivery channel of PDF PIN mailer. Default: Email	Y	
PDF Password Delivery Channel	Delivery channel of PDF password. For security reasons, this channel should be different from PDF Pin Mailer Delivery Channel . Default: SMS	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Pin Mailer Workflow			
PDF Pin Mailer Workflow Type	Refer to Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer . It supports Single-Control and Dual-Control PIN mailers. Select the type of PIN mailer workflow from the drop-down list. Default: Dual-Control PIN Mailer	Y	This setting is mandatory provided the Allow Reset by System Assigned PDF (Pending) option is allowed.

Scripted Rules Tab

The SafeNet PHEFT login module has the **Scripted Rules** tab, which is hidden by default. To enable this tab:

1. Navigate to `<AM5_ROOT>/tomcat/webapps/am5/WEB-INF/classes/` and open the **wpe_config.properties** file.
2. Enable the **Scripted Rules** advanced feature in the admin console. To do so, first search for the `advancedFeatures.enable` element and change the associated value to "true":

```
Properties (.properties files)

advancedFeatures.enable =true
```

3. Add the line `loginModule.scriptedRules = true` to the file:

```
Properties (.properties files)

advancedFeatures.enable =true
loginModule.scriptedRules =true
```

4. Save the changes and restart the AccessMatrix server.

In the **Scripted Rules** tab, the user can define scripted rules to set the login account end time for passwords reset via system-assigned PDF. Example:

Code

```
if (event.isAfterResetPassword() &&
event.isResetPasswordGenerationOptionSystemPDF())
{ action.setLoginAccountEndTimeInHours(24); }
```

Reset the user's password and select the **System Assigned PDF** option. The login module end time will be updated.

Configure SafeNet ProtectServer E2EEA Login Module

In supporting E2EEA with PDF PIN mailer for SafeNet PSE 2 HSM, you must create a new login module. This login module provides the SafeNet ProtectServer E2EEA login method for the user who may reside in either an internal or external user store.

To create a SafeNet ProtectServer E2EEA login module:

1. Navigate to **Administration Dashboard > Configuration > Authentication** and click **Login Module** on the left pane.
2. Click the **Create** button and select the "SafeNet ProtectServer E2EEA Login Module" implementation from the **Choose Login Module Implementation** page.
3. Then, click **Next** and provide the necessary values to the **Configuration** tab attributes (refer to [UAS E2EEA Developer Guide - UAS E2EEA with SafeNet ProtectServer HSM Implementation](#) for the configuration details).
4. Select the **PDF Pin Mailer** tab and provide the necessary values for the PIN mailer attributes before saving the changes.

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Tab			
Reset Password Options			
Allow Reset By System Assigned PDF	Set to "true" to automatically generate the password in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Allow Reset By System Assigned PDF (Pending)	Set to "true" to reset the password using PIN mailer workflow, where upon completion, the password is automatically generated in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
Allow Send PDF Password in Response	Set to "true" to allow the PDF password to be sent in the response to the Password Reset REST API call.	Y	N
Password Generation Policy (The Settings Below Must Comply With the Password Policies)			
Notes: - The settings below must comply with the password policy configured under the Password Quality Policy attribute (or, the Password Quality Policy (System Assigned) attribute instead, if it is configured) in the Configuration tab of the login module. - In the case of conflicting settings within the configured password quality policy, the following attributes take a higher priority when generating the password: Password should not contain these characters Password should not begin with any of these characters Password should not end with any of these characters Password must contain at least one of these characters Refer to AccessMatrix Common Administration Guide - Configure Password Quality Policy for more information on these password quality policy attributes.			
Password Length	Length of the generated password. For the numeric type, the maximum length is 10. Value should be from 6 to 30. Default: 6	Y	N
Password Type	Type of characters in a generated password. Default: Uppercase Alphanumeric	Y	N
PDF Pin Mailer Generation Configuration			
PDF Signing Key	Use a certificate with a private key to sign PDF. New keys can be created at Administration Dashboard > Security	N	N

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	Policy > Key Management > Encryption Key.		
PDF Pin Mailer Validity Period (days)	This value is used to calculate the expiry date of the PDF PIN mailer where PDF mailer expiry date = creation date of the PDF PIN mailer + PDF Pin Mailer Validity Period. Note: To show the expiry date in PDF, set the <<expiryDate>> macro in the PDF Pin Mailer Message Template.	Y	N
PDF Pin Mailer Expiry Date Format	Date format of the PDF PIN mailer expiry date. Symbol Meaning [Example] y year [e.g. yyyy for 2012] M month in year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9] d day in month [e.g. dd for 10] z time zone [e.g. z for PST] Example: To display expiry date 10/11/2025 in PDF, enter dd/MM/yyyy.	Y	
Message Template Configuration			
PDF Pin Mailer Message Template	Message template of the PDF PIN mailer ("Coordinate-based Template") which allows you to put information like a PIN value in an encrypted PDF. It can be created at Administration Dashboard > Configuration > Messaging & Response, select "Coordinate-based Template" for the implementation. Set the following in the message template body: • <<E2EEPIN>> to specify the PIN value in PDF • \${pinmailer.xxx} to specify attribute passed from API(using pinmailer.attr.xxx in the params) • \${userAttr.email} to specify the email of the user if the user has its email attribute stored or mapped in AccessMatrix user store (e.g.	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	LDAP, defaultstore or JDBC user store).		
PDF Delivery Message Template	Message template to include the PDF PIN mailer; this template allows you to attach the PDF file in email. It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to attach a PDF file (i.e. "Default Template Engine"). You must set Allow File Attachment to "true" in the message template.	Y	
PDF Password Message Template	Message template of PDF Password allows you to send the PDF password (PDF password is required to open the encrypted PDF file). It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to send a text (i.e. "Default Template Engine"). Set the following in the message template: • \${pdfpassword} in the body field to specify the PDF password. • \${targetUserEmail} in the recipient field to specify user's email. • \${targetUserMobile} in the recipient field to specify the user's mobile.	Y	
Delivery Channel Configuration			
PDF Pin Mailer Delivery Channel	Delivery channel of PDF PIN mailer. Default: Email	Y	
PDF Password	Delivery channel of PDF password. For security reasons, this channel should be different from the PDF Pin Mailer	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Delivery Channel	Delivery Channel. Default: SMS		
Pin Mailer Workflow			
PDF Pin Mailer Workflow Type	Refer to Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer . It supports Single-Control and Dual-Control PIN mailers. Select the type of PIN mailer workflow from the drop-down list. Default: Dual-Control PIN Mailer	Y	This setting is mandatory provided Allow Reset by System Assigned PDF (Pending) option is allowed.

Configure Thales payShield Login Module

In supporting E2EEA with PDF PIN mailer for Thales payShield HSM, you must create a new login module. This login module provides the Thales payShield login method for the user who may reside in either an internal or external user store.

To create a Thales payShield login module:

1. Navigate to **Administration Dashboard > Configuration > Authentication** and click **Login Module** on the left pane.
2. Click the **Create** button and select the "Thales payShield Login Module" implementation from the **Choose Login Module Implementation** page.
3. Then, click **Next** and provide the necessary values to the **Configuration** tab attributes (refer to [UAS E2EEA Developer Guide - UAS E2EEA with Thales payShield 10K HSM Implementation](#) for the configuration details).
4. Select the **PDF Pin Mailer** tab and provide the necessary values for the PIN mailer attributes before saving the changes.

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Tab			

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Reset Password Options			
Password Type	Numeric.		
Allow Reset By System Assigned PDF	Set to "true" to automatically generate the password in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
Allow Reset By System Assigned PDF (Pending)	Set to "true" to reset the password using PIN mailer workflow, where upon completion, the password is automatically generated in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
PDF Pin Mailer Generation Configuration			
PDF Signing Key	Use a certificate with a private key to sign PDF. New keys can be created at Administration Dashboard > Security Policy > Key Management > Encryption Key.	N	N
PDF Pin Mailer Validity Period (days)	This value is used to calculate the expiry date of the PDF PIN mailer where PDF mailer expiry date = created date of the PDF PIN mailer + PDF Pin Mailer Validity Period. Note: To show the expiry date in PDF, set <<expiryDate>> macro in the PDF Pin Mailer Message Template.	Y	N
PDF Pin Mailer Expiry Date Format	Date format of the PDF PIN mailer expiry date. Symbol Meaning [Example] y year [e.g. yyyy for 2012] M month in year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9] d day in month [e.g. dd for 10] z time zone [e.g. z for PST]	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	Example: to display expiry date 10/11/2025 in PDF, enter dd/MM/yyyy.		
Message Template Configuration			
PDF Pin Mailer Message Template	Message template of the PDF PIN mailer ("Coordinate-based Template") which allows you to put information like a PIN value in an encrypted PDF. It can be created at Administration Dashboard > Configuration > Messaging & Response - select "Coordinate-based Template" for the implementation. Set the following in the message template body: • <<E2EEPIN>> to specify the PIN value in PDF • \${pinmailer.xxx} to specify attribute passed from API (using pinmailer.attr.xxx in the params) • \${userAttr.email} to specify the email of the user if the user has its email attribute stored or mapped in AccessMatrix user store (e.g. LDAP, defaultstore or JDBC user store).	Y	
PDF Delivery Message Template	Message template to include the PDF PIN mailer; this template allows you to attach the PDF file in email. It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to attach a PDF file (i.e. "Default Template Engine"). You must set Allow File Attachment to "true" in the message template.	Y	
PDF Password Message Template	Message template of PDF Password allows you to send the PDF password (PDF password is required to open the encrypted PDF file). It can be created at Administration Dashboard >	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to send a text (i.e. "Default Template Engine"). Set the following in the message template: • \${pdfpassword} in the body field to specify the PDF password. • \${targetUserEmail} in the recipient field to specify user's email. • \${targetUserMobile} in the recipient field to specify the user's mobile.		
Delivery Channel Configuration			
PDF Pin Mailer Delivery Channel	Delivery channel of the PDF PIN mailer. Default: Email	Y	
PDF Password Delivery Channel	Delivery channel of PDF password. For security reasons, this channel should be different from the PDF Pin Mailer Delivery Channel . Default: SMS	Y	
Pin Mailer Workflow			
PDF Pin Mailer Workflow Type	Refer to Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer . It supports Single-Control and Dual-Control PIN mailers. Select the type of PIN mailer workflow from the drop-down list. Default: Dual-Control PIN Mailer	Y	This setting is mandatory provided the Allow Reset by System Assigned PDF (Pending) option is allowed.

Configure Utimaco Crypto Server E2EEA Login Module

In supporting E2EEA with PDF PIN mailer for Utimaco HSM or Crypto Server, you must create a new login module. This login module provides the Utimaco Crypto Server E2EEA login method for the user who may reside in either an internal or external user store.

To create a Utimaco Crypto Server E2EEA login module:

1. Navigate to **Administration Dashboard > Configuration > Authentication** and click **Login Module** on the left pane.
2. Click the **Create** button and select the "Utimaco Crypto Server E2EEA Login Module" implementation from the **Choose Login Module Implementation** page.
3. Then, click **Next** and provide the necessary values to the **Configuration** tab attributes (refer to [UAS E2EEA Developer Guide - UAS E2EEA with Utimaco Crypto Server HSM Implementation](#) for the configuration details).
4. Select the **PDF Pin Mailer** tab and provide the necessary values for the PIN mailer attributes before saving the changes.

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
PDF Pin Mailer Tab			
Reset Password Options			
Allow Reset By System Assigned PDF	Set to "true" to automatically generate the password in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
Allow Reset By System Assigned PDF (Pending)	Set to "true" to reset the password using PIN mailer workflow, where upon completion, the password is automatically generated in the encrypted PDF for sending to the user. The password and the encrypted PDF will be generated within UAS. Default: false	Y	N
Allow Send PDF Password in Response	Set to "true" to allow the PDF password to be sent in the response to the Password Reset REST API call.	Y	N
Password Generation Policy (The Settings Below Must Comply With the Password Policies)			

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Note: The settings below must comply with the password policy configured under the Password Quality Policy in the Configuration tab of the login module.			
Password Length	Length of the generated password. For the numeric type, the maximum length is 10. Value should be from 6 to 30. Default: 6	Y	N
Password Type	Type of characters in a generated password. Default: Uppercase Alphanumeric	Y	N
PDF Pin Mailer Generation Configuration			
PDF Signing Key	Use a certificate with a private key to sign PDF. New keys can be created at Administration Dashboard > Security Policy > Key Management > Encryption Key .	N	N
PDF Pin Mailer Validity Period (days)	This value is used to calculate the expiry date of the PDF PIN mailer where PDF PIN mailer expiry date = creation date of the PDF PIN mailer + PDF Pin Mailer Validity Period . Note: To show the expiry date in PDF, set <<expiryDate>> macro in the PDF Pin Mailer Message Template .	Y	N
PDF Pin Mailer Expiry Date Format	Date format of the PDF PIN mailer expiry date. Symbol Meaning [Example] y year [e.g. yyyy for 2012] M month in year [e.g. MMMM for September, MMM for Sep, MM for 09 and M for 9] d day in month [e.g. dd for 10] z time zone [e.g. z for PST] Example: to display expiry date 10/11/2025 in PDF, enter dd/MM/yyyy.	Y	
Message Template Configuration			
PDF Pin Mailer	Message template of the PDF PIN mailer ("Coordinate-based Template") which	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
Message Template	allows you to put information like a PIN value in an encrypted PDF. It can be created at Administration Dashboard > Configuration > Messaging & Response - select "Coordinate-based Template" for the implementation. Set the following in the message template body: • <<E2EEPIN>> to specify the PIN value in PDF • \${pinmailer.xxx} to specify attribute passed from API(using pinmailer.attr.xxx in the params) • \${userAttr.email} to specify the email of the user if the user has its email attribute stored or mapped in AccessMatrix user store (e.g. LDAP, defaultstore or JDBC user store).		
PDF Delivery Message Template	Message template to include the PDF PIN mailer; this template allows you to attach the PDF file in email. It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to attach a PDF file (i.e. "Default Template Engine"). You must set Allow File Attachment to "true" in the message template.	Y	
PDF Password Message Template	Message template of PDF Password allows you to send the PDF password (PDF password is required to open the encrypted PDF file). It can be created at Administration Dashboard > Configuration > Messaging & Response > Message Template. Select a message template implementation, which allows you to send a text (i.e. "Default Template Engine"). Set the following in the message template:	Y	

Attribute Name	Description	For Password Generation Only (Y/N)	Mandatory (Y/N)
	• \${pdfpassword} in the body field to specify the PDF password. • \${targetUserEmail} in the recipient field to specify user's email. • \${targetUserMobile} in the recipient field to specify the user's mobile.		
Delivery Channel Configuration			
PDF Pin Mailer Delivery Channel	Delivery channel of PDF PIN mailer. Default: Email	Y	
PDF Password Delivery Channel	Delivery channel of PDF password. For security reasons, this channel should be different from the PDF Pin Mailer Delivery Channel . Default: SMS	Y	
Pin Mailer Workflow			
PDF Pin Mailer Workflow Type	Refer to Supported Single-Control and Dual-Control Workflow Types for PDF PIN Mailer . It supports Single-Control and Dual-Control PIN mailers. Select the type of PIN mailer workflow from the drop-down list. Default: Dual-Control PIN Mailer	Y	This setting is mandatory provided Allow Reset by System Assigned PDF (Pending) option is allowed.

Configure Authentication Realm

1. Navigate to **Administration Dashboard > Configuration > Authentication > Authentication Realm**.
2. Click **Create**.
3. Enter an **Authentication Realm Id** (e.g. "PDFPinR") and an **Authentication Realm Name** (e.g. "PDF PIN Realm").
4. Specify the **User Lookup Module** value used to validate user identity, e.g. "SystemLookup".

-
5. Specify the **First(1st) Login Module** that will be used for user authentication, e.g. "PHEFT Login Module (PHEFTLM)".
 6. Click **Save**. A new authentication realm is created.

Assign Authentication Realm to User

A user may belong to a default store or an external user store. You may refer to [AccessMatrix Common Administration Guide - Manager User Stores](#) to know about the different types of user stores and how to configure them.

To create a new user in default store, navigate to **Operation Dashboard > User & Segment > User** and click **Create** on the right pane. Provide values to the following attributes that are necessary for PDF PIN mailer implementation before saving the changes.

Attribute Name	Description
*User Id	Unique user Id, e.g. "user01".
*User Name	Name of user, e.g. 'user01'.
*Interactive	Set to "Yes" to make it an interactive user.
User Store	Name of user store the user belongs to, i.e. "Default Store".
Login Accounts Tab	
This tab contains the authentication realm and login module assigned to user. Click Add to display the Choose Authentication Realm dialog box. Select the authentication realm created in previous section (e.g. "PDF PIN Realm (PDFPinR)") and click OK . Both authentication realm and the login module(s) associated to it will be assigned to the user.	

Set **Force Change Password** to "Yes" for the login module. Then, click **Save**. A new user is created.

You may refer to [AccessMatrix Common Administration Guide - Manager Users](#) for detailed information about users.



Note: Click **Reset Password on the View User** page to check if the PIN mailer's configurations with digital signature work accordingly. On the **Reset Password** dialog box, toggle to the **System Assigned PDF** option to automatically generate a new password to a PDF PIN mailer. Check to see if there is a digital signature attached to the PIN mailer. See [Appendix B](#).

UAS PDF PIN Mailer Implementation using API

Refer to [UAS E2EEA Developer Guide - Reset Password](#) for details regarding the API use cases and sample codes of the PDF PIN mailer solution.

4. UAS PDF PIN Mailer Implementation Using API

Refer to [UAS E2EEA Developer Guide - Reset Password](#) for details regarding the API use cases and sample codes of the PDF PIN mailer solution.

5. Appendices

A. Sample Pin Mailer Template



Dear _____,

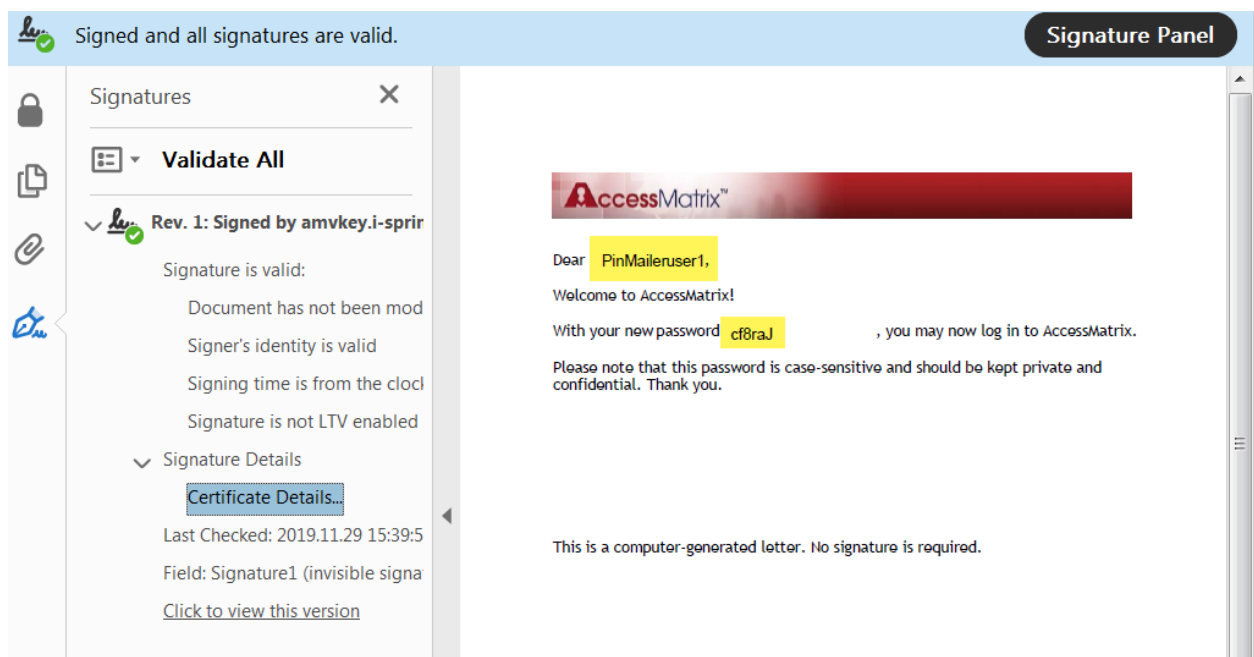
Welcome to AccessMatrix!

With your new password _____, you may now log in to AccessMatrix.

Please note that this password is case-sensitive and should be kept private and confidential. Thank you.

This is a computer-generated letter. No signature is required.

B. Digital Signature Sample

A screenshot of a digital signature verification interface. On the left is a 'Signatures' panel with a lock icon, document icon, and a signature icon. It shows 'Rev. 1: Signed by amvkey.i-sprir' with a green checkmark. Below this, it states 'Signature is valid:' followed by four bullet points: 'Document has not been mod', 'Signer's identity is valid', 'Signing time is from the clock', and 'Signature is not LTV enabled'. Under 'Signature Details', there is a 'Certificate Details...' button and the text 'Last Checked: 2019.11.29 15:39:5' and 'Field: Signature1 (invisible signa'. At the bottom of the panel is a link 'Click to view this version'. The main area on the right is titled 'Signature Panel' and contains the same AccessMatrix banner and text as in section A, but with the name 'PinMaileruser1,' and password 'cf8raJ' highlighted in yellow. The text 'This is a computer-generated letter. No signature is required.' is at the bottom of the main area.

